

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 6: Physical and Chemical Changes

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	6 — Physical and Chemical Changes
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This hardest paper layers several traps at once: a single clue (heat, light, colour, gas, irreversibility, dissolving) is never decisive; mass of the solid product is not the same as conservation of total mass; one event can be physical and chemical together; controlled rust set-ups isolate exactly one factor, so read which factor each pair actually changes; galvanising's sacrificial action is chemical protection, not mere sealing; crystallisation, dissolving, magnetising and state-changes stay physical no matter how dramatic; and counting 'clues' or classifying lists requires checking every member, not the first. Do not write on this paper — mark answers on your answer sheet.

1. A change is observed in four stages: first a clear liquid warms slightly, then it turns from pale yellow to deep orange, then a fine white solid drops out, and finally the original yellow liquid cannot be got back by evaporating. A student asks which one of these four observations, taken entirely alone, would still leave room for doubt that a new substance had formed. Which is it?

- (A) the colour shifting from pale yellow to deep orange, because a colour change can only happen when a new substance forms
- (B) the slight warming, because both physical and chemical changes can release a little heat, so warmth on its own settles nothing
- (C) the fine white solid dropping out, because an insoluble solid appearing from a clear liquid can only be a new substance
- (D) the failure to recover the yellow liquid by evaporating, because that test can sometimes pass even when nothing new formed

2. A 4 g coil of magnesium ribbon is burnt completely; the white magnesium oxide collected weighs 6.6 g. In a separate tray the burning is repeated but this time the rising white smoke is trapped and weighed too, giving 0.4 g of escaped powder. Which single statement about masses is correct?

- (A) the powder gained 2.6 g because burning created new matter from the flame's energy, and the escaped smoke proves some of that new matter floated away
- (B) since the metal was only 4 g, any product over 4 g must include weighing errors, so the true oxide mass is really 4 g
- (C) the 0.4 g of escaped smoke shows magnesium was lost as smoke, so the reaction destroyed 0.4 g of magnesium permanently
- (D) the 2.6 g gain in the collected powder is oxygen that joined from the air, and the 0.4 g of escaped smoke means part of the oxide was simply carried off, not that matter was destroyed

3. Five changes are listed: (i) a wax candle melting, (ii) the same wax vapour burning at the wick, (iii) ice subliming inside a freezer, (iv) milk curdling, (v) a copper wire being drawn thinner. How many of these five are chemical changes?

- (A) three — the burning, the curdling, and the subliming, because subliming changes the solid straight into a different state
- (B) two — the wax vapour burning and the milk curdling, because only those two form new substances
- (C) one — only the burning, because curdling milk merely thickens the same milk without making anything new
- (D) four — every change except drawing the wire, because all of those involve heat or spoiling

4. A controlled rust study uses four sealed tubes, each with one identical nail: T1 = freshly boiled water sealed under oil; T2 = unboiled tap water open to air; T3 = a few drops of water in air sealed with a drying agent removed; T4 = dry air with a drying agent and no water. After two weeks only T2 shows heavy rust. To prove specifically that air (oxygen) is required while water is present, which single pair of tubes should be compared?

- (A) T2 versus T4, because one is wet and open while the other is dry and sealed, which directly shows what rusting needs
- (B) T1 versus T4, because neither rusts, and comparing two tubes that both stay clean confirms air is needed
- (C) T1 versus T2, because both contain liquid water yet only T2 lets air reach it, so they differ only in the presence of air
- (D) T3 versus T4, because both keep a drying agent and so isolate the effect of oxygen cleanly

5. A galvanised steel sheet and a tin-plated steel can are both scratched deeply so the bare steel shows. Left in damp air, the scratch on the galvanised sheet stays rust-free for a long

time, but the scratch on the tin can rusts quickly and even worse than bare steel would. The best explanation within Class-7 ideas is:

- (A) both coatings only seal the surface, so a deep scratch must let both rust at exactly the same rate
- (B) zinc and tin both turn the steel into stainless steel, but the tin layer was simply too thin to finish the job
- (C) zinc is more reactive than iron and corrodes in its place even at the scratch, while tin is less reactive, so at a scratch the iron corrodes instead and the tin can even speed the local rusting
- (D) tin attracts more moisture than zinc, so the difference is just that the tin can stay wetter at the scratch

6. Hot saturated potash-alum solution is split: portion A is cooled very slowly in a still beaker, portion B is cooled quickly by standing the beaker in iced water, and portion C is left in an open dish in the warm sun. All three yield solid alum. Which statement is fully correct?

- (A) cooling is physical but the sun-drying is chemical, because the sun's heat changes alum into a new substance
- (B) all three are chemical, because solid alum is a different substance from alum dissolved in water
- (C) all three are physical (crystallisation or evaporation of the same alum); slow cooling in A tends to give the largest, purest crystals, fast cooling in B gives many tiny ones, and sun-drying in C can trap impurities
- (D) only the slowly cooled portion is physical; the fast-cooled and sun-dried portions form new substances because the change was hurried

7. A teacher passes exhaled breath through fresh limewater until it is milky, then keeps blowing hard for several minutes until the milkiness clears. A student claims the clearing 'proves the whole thing was only physical bubbling.' The best verdict is:

- (A) yes — the milkiness and its clearing are just bubbles of breath scattering and then escaping, a purely physical effect throughout
- (B) no — the milkiness is chemical, but the clearing afterwards is purely physical because the bubbles simply rise out of the liquid
- (C) yes — the clearing shows the calcium carbonate evaporated away, proving the milky stage was only a temporary physical mist
- (D) no — the milkiness was an insoluble new solid (calcium carbonate) forming, a chemical change; the later clearing is excess carbon dioxide turning that solid into a soluble substance, still chemistry, not mere bubbling

8. An iron nail is left in blue copper sulphate solution overnight. Afterwards the nail is reddish-brown and heavier, and the liquid is pale green. A student measures and finds the nail gained 0.64 g while the liquid would, if dried, give a different solid than before. Which set of statements is fully correct?

- (A) the nail rusted, the brown coat is rust, and the green is copper that leaked out of the solution, so the change is just physical mixing
- (B) the nail lost iron and got lighter, and the green is the old copper sulphate merely fading, so nothing chemical happened
- (C) the nail gained a coat of copper metal (so it is heavier and reddish-brown), the green liquid is iron sulphate, and the change is chemical because the more reactive iron displaced copper
- (D) the iron simply dissolved like sugar in the blue liquid, and both the brown coat and the green colour are tricks of the lighting

9. Three claims are made about telling change types apart. (P) 'If a change is reversible it must be physical.' (Q) 'If a change is irreversible it must be chemical.' (R) 'If the starting material can be recovered by a physical method, no new substance formed.' Which claims are actually trustworthy?

- (A) all three are trustworthy, because reversibility and recovery always line up with the type of change
- (B) only Q is trustworthy, because anything that cannot be undone must have made a new substance
- (C) only R is trustworthy; P is usually true but has exceptions, and Q is false because a shattered pot is irreversible yet physical
- (D) only P and Q together are trustworthy, because reversibility decides the type in both directions

10. Two students set up rusting tests. Aria wants to show moisture is needed while keeping air available; Dev wants to show air (oxygen) is needed while keeping water available. Which single pairing gives each of them the correct experiment?

- (A) Aria: a nail in salty water versus a nail in plain water; Dev: a nail painted versus a nail left bare, since each pair changes just one thing
- (B) Aria: a nail in boiled water under oil versus a nail in dry air; Dev: a nail in moist air versus a nail in dry air, since both pairs isolate their factor
- (C) Aria: a nail in open dry air with a drying agent versus a nail in open moist air; Dev: a nail in boiled water sealed under oil versus a nail in plain water open to air
- (D) Aria: a nail in dry air versus a nail in boiled water under oil; Dev: a nail in salty water versus a nail in plain water, since salt supplies the oxygen

11. A laboratory note records six observations from one experiment: the beaker warmed, a fizzing gas escaped, a sharp new smell arose, a fine solid settled out, the mixture changed colour, and the starting liquid could not be recovered by evaporating. How many of these six are recognised clues that a chemical change occurred?

- (A) four, because warmth and the failure to recover the original are not really clues of a chemical change
- (B) three, counting only the gas, the precipitate, and the colour change, since smells and heat mean nothing
- (C) five, because a colour change can happen physically and so should not be counted as a clue
- (D) all six, because warmth, a gas, a new smell, a precipitate, a colour change, and inability to recover the original by a physical step are each recognised clues

12. A blacksmith heats an iron rod until it glows orange, hammers it into a hook, quenches it in water, and a thin grey-black flaky scale forms on the surface; beneath the scale the metal is still iron. Considering three events — (i) the rod glowing, (ii) the rod being bent into shape, (iii) the grey-black scale forming — the correct classification is:

- (A) all three are chemical, because heating iron until it glows always turns it into a new substance
- (B) all three are physical, because the bulk of the rod stays iron throughout
- (C) (i) physical (hot iron giving off light), (ii) physical (only the shape changes), and (iii) chemical (iron combined with oxygen to make a new scale)
- (D) (i) chemical and (ii) chemical because the glow and the hammering both force a reaction, but (iii) physical because scale just flakes off

13. Rock salt contaminated with insoluble clay and a soluble coloured impurity must be purified to pure white salt crystals. Which ordered procedure is correct?

- (A) heat the rock salt strongly until the clay and the coloured impurity both burn away, then collect the white salt left behind
- (B) grind the rock salt and pick the white grains from the clay by hand, since the soluble impurity will fall off as dust
- (C) dissolve the rock salt, crystallise it at once without filtering, and the clay will sink while the coloured impurity rises off as gas
- (D) dissolve the rock salt in water, filter off the clay, then slowly crystallise so pure salt comes out while the soluble coloured impurity stays behind in the leftover solution

14. Four iron objects are stored for a year with intact coatings: W = wrapped in greased cloth indoors; X = painted and kept outdoors; Y = zinc-galvanised and kept outdoors; Z = bare and kept in a humid shed. Rank them from LEAST rusted to MOST rusted.

- (A) Z least, then X, then Y, then W most, because grease attracts moisture and so rusts the iron fastest of all
- (B) W and Y least, then X, then Z most — grease, intact zinc, and paint all keep the iron protected, while the bare object in humid air rusts freely

- (C) Y most rusted, because its zinc coating reacts with the air and corrodes faster than the bare iron does
- (D) all four rust the same amount, because once a year passes every iron object rusts equally regardless of coating

15. A bright copper coil is heated in a flame and turns black; scraped, fresh copper shows beneath. Separately, a polished silver spoon left in air for months slowly darkens (tarnishes). A student says 'both are just the metals getting hot.' The best classification is:

- (A) both are chemical surface changes — the copper combined with oxygen to make a black compound, and the silver combined with substances in the air to form a dark tarnish
- (B) both are physical, because the copper and the silver are still metal underneath in each case
- (C) the copper is chemical but the silver tarnishing is physical, because tarnishing happens slowly without a flame
- (D) both are just the colour of hot metal, since heating any metal makes its surface look darker

16. A student argues: 'Any change that takes IN heat (feels cold) cannot be chemical, because chemical changes always give OUT heat.' Which single example best refutes this rule?

- (A) dissolving an anhydrous solid in water so the beaker warms up, then recovering the same solid by evaporating
- (B) melting ice, which takes in heat as it turns to water and freezes back to the same ice on cooling
- (C) heating limestone strongly so it takes in heat and gives off a gas, leaving a new solid — a chemical change that absorbs heat
- (D) burning magnesium ribbon, which gives out heat and dazzling light while forming a white powder

17. Two clear, colourless solutions are mixed; nothing happens for a moment, then a fine white solid forms and a faint smell appears, but the temperature stays unchanged. A student concludes 'no temperature change, so it must be physical.' The correct verdict is:

- (A) a chemical change occurred; a precipitate forming and a new smell appearing are clues of new substances, and a chemical change need not show any temperature change
- (B) a physical change occurred, because without any heat given out or taken in there can be no chemical reaction
- (C) no change of either kind occurred, since the temperature was the only reliable test and it stayed the same
- (D) a physical change occurred, because mixing two colourless liquids can only blend them, never react them

18. A factory buries long iron pipes in damp, salty coastal soil where coatings often get scratched during burial. Three options are weighed: painting only, wrapping in oily cloth, or galvanising with zinc. Which choice gives the most durable protection given the scratching, and why?

- (A) galvanising, because even at a scratch the more reactive zinc keeps corroding in place of the iron, protecting the exposed metal that paint or oil would leave bare
- (B) painting, because paint chemically converts the iron into a rust-proof metal that scratches cannot undo
- (C) oily cloth, because the oil feeds the iron a slow, controlled supply of oxygen so only harmless rust forms
- (D) all three are equal, because each only seals the surface and a scratch ruins every coating in exactly the same way

19. Within Class-7 scope, which single word equation for rusting is the most complete and correct?

- (A) iron + oxygen (from air) + water $\rightarrow$ hydrated iron oxide (rust)
- (B) iron + carbon dioxide + water $\rightarrow$ rust, because the carbon dioxide in the air is what attacks the iron
- (C) iron + nitrogen $\rightarrow$ rust, because nitrogen is by far the most plentiful gas in the air
- (D) iron + salt $\rightarrow$ rust, because salt is the substance that joins with iron to make the brown solid

20. After magnesium ribbon burns, the white powder is stirred into a little water and the liquid turns red litmus paper blue. A student says this just proves 'the magnesium was always able to turn litmus blue.' The correct interpretation is:

- (A) the ribbon was only melted and re-solidified, so it kept all the original properties of magnesium metal
- (B) burning produced a genuinely new substance (magnesium oxide) whose water mixture behaves like a base, and this differs from the metal, confirming a chemical change
- (C) the litmus result comes from leftover grit used to clean the ribbon, not from any new substance at all
- (D) magnesium metal itself turns red litmus blue, so no new substance is needed to explain the colour change

21. Camphor is gently warmed in a dish under a cool inverted funnel; the solid leaves the dish without melting and white camphor reappears on the cool funnel, with no liquid ever seen. Naphthalene balls in a cupboard also slowly shrink and vanish over weeks. How should these be classified?

- (A) both are chemical, because heating broke the camphor down and the naphthalene reacted with cupboard air to disappear
- (B) the camphor is physical but the naphthalene is chemical, because the naphthalene vanishes completely instead of reappearing
- (C) the camphor is chemical (vaporising) followed by a physical step (re-forming), and the naphthalene is purely physical
- (D) both are physical — camphor sublimates then deposits as the same camphor, and the naphthalene balls sublime away, with no new substance formed in either

22. A student lists four 'rules of thumb' and asks which one is FALSE within Class-7 science.

- (A) a chemical change forms one or more new substances, while a physical change forms none
- (B) some physical changes, such as freezing water, release heat to the surroundings
- (C) crystallisation gives pure crystals of the same substance and is a physical change
- (D) every change that gives off a gas is a chemical change

23. An iron gate is painted, but after a year a tiny chip in the paint shows a rust spot that has spread under the surrounding paint and lifted it, exposing more bare iron. Which explanation correctly links the chip to the spreading?

- (A) the paint itself slowly turned into rust across the whole gate at the same time, and the chip just made it visible
- (B) the chip let in extra paint solvent, and it was the solvent, not air and moisture, that corroded the iron
- (C) at the chip, air and moisture reached the bare iron and rust formed; the rust takes up more space and pushes off nearby paint, exposing fresh iron that then rusts in turn
- (D) the rust spot proves that painting actually speeds up rusting compared with leaving the iron bare

24. A solid is heated and slowly turns into a clear liquid of the same material; on cooling it sets back to the identical solid. Of four single observations a student might use to decide the change type, which one would be the WEAKEST proof that a chemical change had occurred?

- (A) a gas with a sharp, completely unfamiliar smell was steadily given off
- (B) a fine insoluble solid suddenly settled out when two clear liquids were combined
- (C) a solid was warmed and turned into a clear liquid of the same material, recoverable on cooling
- (D) the material burst into flame, giving out bright light and intense heat while a residue was left

25. A pale-green iron sulphate solution is left in an open dish for days; a brownish film appears at the surface and slowly thickens. Within Class-7 ideas, the best explanation is:

- (A) the green colour is simply fading in the daylight, a physical change with no new substance formed
- (B) the solution is crystallising at the surface, and those crystals just happen to look brown
- (C) the iron sulphate is dissolving still further, and very concentrated solutions always look brown
- (D) the iron in the solution is reacting with oxygen and moisture from the open air, a chemical change forming a rust-like brown substance

26. Five everyday events are checked. Which one shows BOTH a physical change and a chemical change happening within the SAME single event?

- (A) ice melting in a glass — the ice melting is physical and the drink getting colder is the chemical part
- (B) salt dissolving in water — the salt spreading out is physical and the water becoming salty is the chemical part
- (C) a wire being bent — the bending is physical and the wire warming from the bending is the chemical part
- (D) a candle burning — the solid wax melting is physical, while the wax vapour burning at the flame is chemical

27. A student claims 'cooking is always a chemical change.' Which single kitchen step best shows the claim is too broad because that step is actually physical?

- (A) boiling an egg until the runny white turns solid and opaque
- (B) toasting bread until its surface browns and gives off a roasted smell
- (C) melting butter in a warm pan, where it later sets back into the same butter on cooling
- (D) caramelising sugar in a hot pan until it turns brown and bitter

28. Stainless-steel cutlery and a chrome-plated tap both resist rust for years. A student asks why the iron in each does not rust. The best shared explanation within Class-7 scope is:

- (A) in both, the iron has been chemically turned into pure chromium, which simply cannot rust
- (B) in both, a rust-resistant chromium-containing layer or alloy keeps a protective surface between the iron and the air and moisture
- (C) both resist rust only because the items are always kept perfectly dry and never touch any water
- (D) both make the iron react with oxygen so quickly that the iron is used up before rust can form

29. Four nails are kept identical except for one factor each: N1 in moist air with oxygen present; N2 in moist air with the oxygen removed; N3 in dry air with oxygen present; N4

in dry air with oxygen removed. Only N1 rusts heavily; the others barely rust. The single clearest conclusion is:

- (A) rusting needs only oxygen, since the dry-air nail with oxygen should then have rusted just as heavily
- (B) rusting needs only moisture, since the moist-air nail without oxygen should then have rusted just as heavily
- (C) rusting needs neither oxygen nor moisture, since iron rusts in time no matter how it is kept
- (D) rusting needs BOTH oxygen and moisture together, since rust is heavy only where both are present

30. A change releases heat AND forms a new substance; a second change absorbs heat AND forms a new substance; a third change releases heat but forms NO new substance; a fourth change absorbs heat and forms no new substance. How many of these four are chemical changes?

- (A) one, because only a change that releases heat can possibly be chemical
- (B) two, because forming a new substance is what makes a change chemical, regardless of whether heat is taken in or given out
- (C) three, because any change that involves heat at all is chemical
- (D) four, because forming a new substance and exchanging heat each independently make a change chemical

31. Common salt is dissolved in water; half the solution is dried gently in the sun and half is heated very strongly in a metal dish. Both leave white solid. A student says the strongly heated half 'must be a new substance because so much heat was used.' The correct verdict is:

- (A) the strongly heated half is chemically new, because applying enough heat always creates a new substance
- (B) both halves give back the same salt; using a lot of heat merely evaporates the water faster and is still a physical change, since no new substance forms
- (C) the sun-dried half is physical but the strongly heated half is chemical, because the amount of heat decides the type
- (D) both halves are chemical, because evaporating water away from salt always makes a fresh compound

32. Aman says 'a change is chemical only if you can SEE something happen — a colour, a gas, or a flame.' Bina says 'some chemical changes show no obvious clue at the moment, yet a new substance has still formed.' Who is right and why?

- (A) Bina, because the real test is whether a new substance formed, which can be true even when no dramatic clue is visible at the time
- (B) Aman, because if no clue is visible then no new substance can possibly have formed

(C) Aman, because chemical changes are simply defined as the changes that show a visible signal

(D) neither, because a change that shows no visible clue is always physical by definition

33. A bright iron wire is held in a flame and burns with orange sparks, leaving dark crumbly bits; a second piece of the same wire left in damp salty air slowly turns brown and flaky. A student says the fast fiery one is chemical but the slow one is 'just physical because it is slow.' The best classification is:

(A) both are chemical changes in which iron combines with oxygen — fast and fiery in the flame, slow as rusting in damp salty air

(B) the flame event is chemical but the slow browning is physical, because slow changes are always physical

(C) both are physical, because the wire is still essentially iron in each case

(D) the slow browning is chemical (rusting) but the fiery burning is physical, because sparks are only hot bits flying off

34. A bottle of fresh limewater is left loosely corked for several weeks and a faint white skin forms on its surface. The most likely reason within Class-7 chemistry is:

(A) the water slowly evaporated, leaving a purely physical film of dust on the top

(B) carbon dioxide from the air slowly reacted with the limewater to form a thin layer of insoluble white solid, a chemical change

(C) the limewater froze a little at the surface where the air kept it coolest

(D) oxygen from the air dissolved in and bleached the surface layer white

35. A small lump of solder (a low-melting metal) is melted on a tin lid, cooled to a hard lump, then re-melted, each time the same shiny metal. A friend insists 'melting a metal must be chemical because metals are special.' The correct response is:

(A) yes, melting a metal is chemical, because the heat turns the metal into a new substance each time it melts

(B) melting is physical but the cooling is chemical, because the cooled lump counts as a brand-new solid

(C) it is rusting, because heating any metal on a lid in air always corrodes it into a new substance

(D) melting and re-solidifying the solder are physical state changes; the metal keeps its identity, so being a metal does not make melting chemical

36. A coloured fruit-drink concentrate is bright red; adding water makes it pink, and adding still more water makes it nearly colourless. A student says 'the fading colour proves a chemical reaction with the water.' The correct verdict is:

(A) no chemical change occurred; diluting only spreads the same colouring substance through more water, a physical change recoverable by evaporation

(B) a chemical change occurred, because any colour change is proof that a new substance formed

(C) a chemical change occurred, because adding water always reacts with a dye and weakens it

(D) it cannot be decided, because colour changes can never be classified as physical or chemical

37. A magnet is stroked along a steel needle until the needle can pick up pins; later the needle is heated strongly and dropped, and it loses its magnetism. Throughout, the needle stays steel. Magnetising and de-magnetising the needle are:

(A) both chemical changes, because the needle gains and then loses magnetic energy

(B) a chemical change (magnetising) followed by a physical change (de-magnetising)

(C) rusting, because strongly heating steel in air always corrodes it into a new substance

(D) both physical changes, because the needle remains the same steel and no new substance forms in either step

38. Equal spoons of common salt are stirred into (i) plain water and (ii) water that already has copper sulphate dissolved in it. In both the salt simply dissolves and nothing new appears. A student claims 'in the copper sulphate cup the two dissolved substances must have reacted.' The correct conclusion is:

(A) in the copper sulphate cup a chemical reaction occurred, because two different dissolved substances always react

(B) the salt was destroyed in both cups, because dissolved salt can never afterwards be recovered

(C) in both cups the salt only dissolved, a physical change; dissolving makes no new substance, so no reaction occurred in either

(D) dissolving in the copper sulphate water is chemical while in plain water it is physical, because the colours differ

39. Two identical iron benches are installed on the same day: one on a breezy beachfront promenade and one in a dry inland courtyard. After one monsoon the beachfront bench is far rustier. A student offers four reasons; which is the best within Class-7 ideas?

(A) the sea breeze keeps the beachfront bench cooler, and cooler iron is the kind that rusts the fastest

(B) the salty, humid sea air supplies both extra moisture and dissolved salt, and salt speeds up rusting, so the beachfront bench corrodes faster than the dry inland one

(C) the beachfront air is richer in carbon dioxide, and carbon dioxide is the gas that actually combines with iron to make rust

(D) the inland bench was cast from a stronger grade of iron that resists rust even when soaked, while the beach bench is ordinary iron

40. A change is studied with four tests: (1) tried to recover the starting material by a physical means — could not; (2) measured the temperature — it rose a little; (3) looked for a colour change — none; (4) checked for a gas — none. Which single result is the strongest evidence the change is chemical?

- (A) Test 2 — the temperature rose, which by itself proves a chemical change
- (B) Test 3 — no colour change was seen, which by itself proves the change must be chemical
- (C) Test 4 — no gas was seen, which by itself proves a new substance formed
- (D) Test 1 — the starting material could not be recovered by a physical method, indicating a new substance formed

41. Two changes are compared: (i) iced water left out warms up and the ice melts; (ii) a slice of bread left out goes mouldy and develops a bad smell. Choose the correct classification and the reason that distinguishes them.

- (A) both physical, because in each case something is simply left out in the open air to change on its own
- (B) both chemical, because both changes occur because of warmth from the surroundings
- (C) (i) chemical because heat was absorbed, and (ii) physical because the bread merely changed its appearance
- (D) (i) physical, because the ice only changes state to the same water; (ii) chemical, because mould growth produces new substances and a new smell

42. Why does crystallisation give purer copper sulphate crystals than the impure powder it started from? Choose the correct mechanism.

- (A) the soluble copper sulphate dissolves and re-forms as crystals while the insoluble impurities are filtered off and any soluble impurities stay behind in the leftover solution
- (B) the impurities chemically react with the copper sulphate and are turned into more copper sulphate
- (C) the heat used to dissolve the powder burns all the impurities away completely before the crystals form
- (D) the crystals are a brand-new compound that no longer contains any of the original impurities

43. A change releases a gas that, when bubbled into limewater, turns it milky; the mixture also cannot be returned to its starting materials by any simple physical step. A careful student wants the single most decisive reason to call the change chemical. The best answer is:

- (A) the original cannot be recovered by a physical means, showing a new substance formed; the limewater test confirms the gas is carbon dioxide, a product of the new chemistry
- (B) the release of any gas alone proves it, because only chemical changes can ever give off a gas

(C) the limewater turning milky alone proves it, because limewater only goes milky for chemical reactions of any kind

(D) the three clues together still cannot decide it; only measuring the exact heat released can confirm chemistry

44. Four iron items face a year outdoors with their coatings staying intact: a gate painted glossy, a pail galvanised with zinc, a hinge wiped with oil, and a bracket left completely bare. A student predicts the bare bracket and the oiled hinge will rust the most. Why is this prediction wrong?

(A) the prediction is right, because oil soaks into iron and draws in moisture that rusts it fastest of all

(B) the prediction is wrong only about the gate, because glossy paint actually speeds rusting more than bare iron

(C) the oiled hinge is protected because the oil keeps air and moisture off the iron, so only the bare bracket rusts heavily; oil prevents rather than promotes rusting

(D) the prediction is right, because the galvanised pail rusts even faster than the bare bracket as its zinc reacts away

45. Which single pair correctly classifies BOTH of its members?

(A) galvanising an iron sheet (the coating step) — chemical; cooking an egg — physical

(B) fermentation of dough by yeast — chemical; sharpening a pencil with a blade — physical

(C) sublimation of naphthalene balls — chemical; rusting of a gate — physical

(D) dissolving washing soda in water — chemical; digesting a meal — physical

46. A teacher mixes lemon juice with baking soda; it fizzes vigorously, and the escaping gas turns limewater milky and does not let a burning splint keep burning. From this, the correct conclusions are:

(A) a chemical change occurred and the gas is carbon dioxide, shown by the limewater turning milky and the gas not supporting burning

(B) a physical change occurred, because fizzing is just dissolved air escaping as it does from a cold drink

(C) the gas is oxygen, because vigorous bubbling shows a gas that strongly supports burning

(D) the gas is hydrogen, because acids always release hydrogen and hydrogen makes bubbles

47. Which one comparison about energy in physical and chemical changes is correct?

(A) both physical and chemical changes can absorb or release energy, so energy flow alone does not tell you which type a change is

(B) only chemical changes can release energy; physical changes never give out any heat

- (C) only physical changes can absorb energy; chemical changes always release it
- (D) all energy-releasing changes are chemical and all energy-absorbing changes are physical

48. A glass of warm sugar solution is left uncovered for days; clear sugar crystals grow on the glass, but a few black specks appear where sugar at the dry edge overheated and browned. The crystal-forming and the browning are, respectively:

- (A) both physical changes, because it is all sugar coming out of the same solution
- (B) both chemical changes, because the sugar separated itself out from the water in each case
- (C) a physical change (crystallisation of the same sugar) and a chemical change (sugar browning into a new substance)
- (D) a chemical change (the crystallising) and a physical change (the browning), because new crystals are a new substance

49. Inside an unopened food can the iron is coated with tin, while a bucket is coated with zinc. A child argues 'tin and zinc protect iron in exactly the same way.' Within Class-7 scope, the most accurate reply is:

- (A) they are identical, because each turns the iron beneath into a brand-new rust-proof metal
- (B) both coat the iron to keep air and moisture away, but zinc, being more reactive than iron, also keeps protecting at a scratch, while tin does not, so they are not identical
- (C) tin makes iron rust faster while zinc stops it, so they work in completely opposite ways with no shared idea
- (D) neither tin nor zinc has anything to do with rusting; only the food inside keeps the can's iron from rusting

50. A student summarises rusting, burning, and respiration. Which comparison is correct?

- (A) only burning is chemical; rusting and respiration are physical because the iron and the body stay the same
- (B) all three are chemical changes in which a substance combines with or uses oxygen – burning is fast, respiration is a slow controlled use inside the body, and rusting is slow surface oxidation
- (C) all three are physical, because none of them makes any new substance
- (D) rusting and burning are chemical, but respiration is physical because it only releases warmth

51. A student writes: 'Because dissolving salt and evaporating it back are both reversible, reversibility goes with physical changes.' Which case best SUPPORTS that reversibility goes with physical changes?

- (A) burning a candle and then trying to collect the smoke back into solid wax
- (B) cooking an egg and then cooling it to get the runny egg back again

- (C) dissolving salt in water and then evaporating to recover the very same salt — reversible, with the same substance throughout
- (D) rusting an iron nail and then drying it to turn the rust back into shiny iron

52. A boiled egg cannot be un-boiled and a smashed clay tile cannot be un-smashed. A student concludes 'both are chemical because both are irreversible.' The verdict for the egg is right but the reasoning is flawed because:

- (A) irreversibility alone does not prove a chemical change; the egg is chemical because new substances form, while the smashed tile is irreversible yet still the same clay material, so physical
- (B) irreversibility always guarantees a chemical change, so the reasoning is actually perfectly fine
- (C) both the boiling and the smashing are in fact physical, so the verdict for the egg is wrong
- (D) neither the boiling nor the smashing forms a new substance, so neither is chemical

53. A controlled experiment finds: a nail in dry air does not rust; a nail in pure boiled water sealed from air does not rust; a nail in ordinary water open to air does rust. A student concludes 'water alone causes rust.' The correct conclusion is:

- (A) water alone causes rust, since the nail in open water clearly rusted
- (B) water alone is not enough; rusting needs both water and air (oxygen), since the sealed boiled-water nail had water but no air and did not rust
- (C) air alone causes rust, since the dry-air nail was exposed to plenty of air
- (D) neither water nor air is needed; the open-water nail rusted only because it was disturbed

54. Exactly one of the following four lists is correctly ordered as physical, chemical, physical. Which list is it?

- (A) dissolving sugar in tea; toasting bread to brown; bending a metal spoon
- (B) rusting a nail; melting candle wax; souring of milk
- (C) freezing cream; crystallising salt; sharpening a pencil
- (D) burning a sparkler; tearing paper; digesting a meal

55. Near a candle three events happen together: solid wax melts, the liquid wax climbs the wick, and wax vapour burns at the flame. The correct count of physical and chemical changes among these three is:

- (A) two physical (the wax melting and the same wax rising up the wick) and one chemical (the wax vapour burning to new substances)
- (B) all three are chemical, because they all happen only because of the flame's heat
- (C) all three are physical, because it is wax throughout until it finally disappears
- (D) one physical (the melting) and two chemical (the rising and the burning), because climbing the wick changes the wax

56. A student insists 'any change that needs heat to make it happen is a chemical change.' Which single example best refutes this rule?

- (A) heating sugar until it caramelises into a brown, bitter substance
- (B) heating ice until it melts to water, which freezes back to the same ice on cooling
- (C) heating limestone strongly until it gives off a gas and leaves a new solid
- (D) heating magnesium ribbon until it burns to a white powder

57. Bright copper wire is heated until it is black on the surface; when scraped, fresh copper shows beneath. The black coating is best explained as:

- (A) soot from the flame settling on the wire, which is only physically stuck on
- (B) a new substance (a copper-oxygen compound) formed where copper combined with oxygen in the air, a chemical change at the surface
- (C) the copper merely becoming hot, so the black is just the colour of hot copper
- (D) the copper melting and re-freezing as a black solid, a physical change of state

58. A pure sample of salt is dissolved in water; half is dried in the sun and gives white salt, while the other half is dried over a strong flame and also gives white solid that is checked and behaves exactly like ordinary salt. A student says the flame-dried solid 'must be chemically new because it took fierce heat.' The correct verdict is:

- (A) the flame-dried half is chemically new, because fierce heat applied to a solution always creates a new substance
- (B) the sun-dried half is physical but the flame-dried half is chemical, since the strength of the heat decides the type
- (C) both halves are chemical, because driving water off a dissolved salt always builds a fresh compound
- (D) both halves recover the same salt; evaporating water with fierce heat is still a physical change because no new substance forms

59. Which single statement comparing crystallisation and rusting is correct?

- (A) both are chemical changes, because each produces a solid that was not there before
- (B) both are physical changes, because in each the metal or the salt stays the same substance
- (C) crystallisation is chemical (new crystals form) while rusting is physical (the iron stays iron beneath the rust)
- (D) crystallisation is a physical change giving back the same substance as pure crystals, while rusting is a chemical change forming a new substance from iron, oxygen, and water

60. A clean iron sheet of mass 50 g is left in damp air for a long time. Brown rust forms and keeps flaking off, and after months the bare sheet that remains weighs only 38 g, yet all the flakes carefully collected weigh 21 g. A student is puzzled that the flakes outweigh the 12 g of iron that disappeared. The best explanation within Class-7 ideas is:

(A) rust is iron combined with oxygen and water, so each bit of iron that turned to rust gained mass, which is why 12 g of lost iron gives more than 12 g of flakes even as the solid sheet keeps thinning

(B) the flakes trapped air bubbles that make them weigh more than the iron they came from, with no real gain in matter

(C) rusting created brand-new matter from nothing, so the flakes can weigh anything and the masses cannot be compared

(D) the sheet must have been impure, since pure iron would have gained weight overall instead of thinning to 38 g

Solutions — Science (Class 7), Chapter 6: Physical and Chemical Changes — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	D	31	B	41	D	51	C
2	D	12	C	22	D	32	A	42	A	52	A
3	B	13	D	23	C	33	A	43	A	53	B
4	C	14	B	24	C	34	B	44	C	54	A
5	C	15	A	25	D	35	D	45	B	55	A
6	C	16	C	26	D	36	A	46	A	56	B
7	D	17	A	27	C	37	D	47	A	57	B
8	C	18	A	28	B	38	C	48	C	58	D
9	C	19	A	29	D	39	B	49	B	59	D
10	C	20	B	30	B	40	D	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Warmth alone is the weakest of the four clues, because dissolving an anhydrous salt or other purely physical events can also warm a beaker without forming any new substance. A colour change here from yellow to orange and an insoluble solid settling from a clear liquid (a precipitate) are genuine signs that something new appeared. Inability to recover the starting liquid by a physical step is in fact strong evidence that a new substance formed, the opposite of leaving doubt. So the temperature rise is the observation that, on its own, decides nothing.

2. (D) Burning magnesium combines the metal with oxygen from the air, so the solid oxide is heavier than the metal by the mass of oxygen added; the 2.6 g increase is that captured oxygen, not new matter conjured from energy. Matter is never created, so the gain cannot be invented mass and is not a mere weighing error. The escaping white smoke is oxide carried away in the air, so it was not destroyed but simply lost from the tray; magnesium itself is not vanishing into nothing. Hence the gain is added oxygen and the smoke is displaced oxide.

3. (B) Melting wax, ice subliming, and drawing copper thinner are all physical: the substance keeps its identity and only its state or shape changes. Wax vapour burning forms new substances such as carbon dioxide and water, and milk curdling produces new substances responsible for the sour taste and solid curd; both are chemical. That makes exactly two chemical changes. Subliming is a state change, not chemical, and curdling genuinely is chemical, so neither one nor three nor four is correct.

4. (C) To isolate the effect of air while keeping water constant, the two tubes compared must both have water and differ only in whether air can reach it. T1 (boiled water under oil) has water but its air is excluded, and T2 (tap water open) has both water and air; their only difference is the air, so this pair isolates air. Comparing T2 with T4 changes two things at once (both water and air), so it cannot isolate air alone. T1 versus T4 compares two non-rusting tubes and differs in water as well, and T3 versus T4 both keep a drying agent so neither truly holds water. Only T1 versus T2 changes air alone.

5. (C) Galvanising works because zinc is more reactive than iron, so even where the coating is broken the zinc keeps corroding in preference to the iron and protects the exposed steel. Tin is less reactive than iron, so at a scratch the iron is the one that corrodes, and the local set-up can make that rusting worse than for bare steel. The coatings therefore do not behave identically, neither turns steel into stainless steel, and the difference is about which metal is more reactive, not about which surface holds more water.

6. (C) Crystallisation and evaporation both leave the very same alum, so all three portions undergo physical changes regardless of how fast or by what means the water is removed. Dissolved alum and solid alum are the same substance in different states, not different substances, and the rate of cooling or the sun's heat does not create anything new. The genuine difference is only in crystal quality: slow cooling grows big pure crystals, fast cooling gives many small ones, and open sun-drying can trap impurities. Hence all three are physical.

7. (D) The milky stage is a genuine chemical change: carbon dioxide in the breath reacts with limewater to make insoluble calcium carbonate, the white cloudiness. Continuing to add excess carbon dioxide does not undo chemistry; it converts that solid into a soluble substance, which is why the liquid clears, so the clearing is also chemical, not just bubbles rising or a solid evaporating away. The student is wrong because both the clouding and the clearing involve new substances forming, not a purely physical mist.

8. (C) Iron is more reactive than copper, so it pushes copper out of the solution: copper metal deposits on the nail, making it reddish-brown and heavier, and the iron that goes into solution forms iron sulphate, which is pale green. New substances form, so the change is chemical, not physical mixing. The brown coat is copper, not rust; the nail gains rather than loses mass; and dissolving like sugar would leave the solution unchanged, which is not what happens. So the displacement description is the correct one.

9. (C) Recovering the original by a purely physical method shows that no new substance was ever formed, so R is a reliable test. Q is false: breaking a clay pot cannot be undone by simple means yet makes no new substance, so it is physical despite being irreversible. P often holds but is not airtight as a rule, since reversibility is only a rough guide and is not the defining feature. Therefore only R can be trusted, and claims that lean on irreversibility or treat reversibility as decisive in both directions are wrong.

10. (C) To isolate moisture while keeping air, Aria must compare two open-air nails that differ only in being dry or moist, which her dry-air-versus-moist-air pair does. To isolate air while keeping water, Dev must compare two wet nails that differ only in whether air reaches them, which the boiled-water-under-oil versus open-water pair achieves. Salty-versus-plain water changes salt not moisture-versus-dry, painted-versus-bare changes the whole sealing not one variable, and salt does not supply oxygen, so the other pairings fail to isolate a single factor.

11. (D) The recognised clues of a chemical change include heat being given out, a gas evolving, a new smell, a precipitate forming, a colour change, and the original being unrecoverable by a physical method. All six listed observations match clues on that list, so all six count. It would be wrong to discard warmth or the recovery failure, both of which are listed clues, and a colour change is itself one of the standard clues. Hence the count is six, with the caution that any single clue alone is not decisive.

12. (C) Glowing is hot iron emitting light, with the metal staying iron, so it is physical; bending changes only the shape, also physical. The grey-black surface scale, however, is iron that has combined with oxygen during heating to form a new oxide, which is a chemical change at the surface even though the bulk stays iron. So two events are physical and one is chemical. Treating the glow or the hammering as chemical is wrong, and dismissing the scale as mere flaking ignores the new substance formed.

13. (D) Dissolving lets the salt and the soluble impurity go into solution while the insoluble clay can be filtered out; slow crystallisation then grows pure salt crystals, leaving the soluble coloured impurity behind in the remaining solution, which is the principle that makes crystallisation purify. Strong heating does not selectively burn clay or a coloured impurity away, hand-picking cannot remove a soluble impurity, and skipping the filtration leaves the clay mixed in while a dissolved impurity does not leave as a gas. So the dissolve-filter-crystallise order is correct.

14. (B) Grease and paint form barriers that keep air and moisture off the iron, and intact zinc both seals and, being more reactive, would protect even at a break, so W, X and Y stay well protected; the bare object Z in humid air has free access to both moisture and oxygen and rusts the most. Grease does not attract moisture to speed rusting, zinc's sacrificial corrosion protects rather than ruins the iron, and coatings clearly do not all rust equally, so the only correct ranking places the protected objects ahead of the bare one.

15. (A) Heated copper combines with oxygen to form a new black compound on its surface, a chemical change, shown by the fresh copper still lying beneath. Silver darkening is also chemical: the silver combines with substances in the air to make a dark tarnish, a new substance, even though it happens slowly and without a flame. The fact that metal remains underneath does not make either change physical, slowness does not make tarnishing physical, and neither is merely the colour of hot metal since the silver was never heated. Both are chemical surface changes.

16. (C) Heating limestone is a chemical change that absorbs heat: it takes in energy and gives off a gas while leaving a new solid, directly showing that a chemical change can take in heat rather than always give it out. Dissolving an anhydrous solid releases heat and is physical, melting ice absorbs heat but is a physical state change, and burning magnesium releases heat, so none of those refutes the claim. Only the heat-absorbing chemical change of limestone breaks the rule that chemical changes must give out heat.

17. (A) A precipitate and a new smell are both recognised clues that new substances formed, so the change is chemical. There is no rule that a chemical change must alter the temperature; many reactions proceed with little or no temperature change, so a constant temperature does not rule out chemistry. Mixing two clear liquids can certainly produce a reaction when an insoluble product forms. Hence the lack of a temperature change is not evidence of a physical change here.

18. (A) Galvanising is best where scratching is likely because zinc is more reactive than iron and goes on sacrificing itself to protect the iron even where the coating is broken, whereas a scratch through paint or oil leaves bare iron exposed to rust. Paint does not chemically turn iron into a rust-proof metal, oil does not feed controlled oxygen to make 'harmless' rust, and the three options are not equal precisely because only zinc keeps protecting at a scratch. So galvanising gives the most durable protection.

19. (A) Rusting needs both oxygen from the air and water, and the brown product is a hydrated iron oxide, so $\text{iron} + \text{oxygen} + \text{water} \rightarrow \text{rust}$ is the complete correct equation. Carbon dioxide is not the gas that rusts iron, nitrogen does not combine with iron to form rust despite being plentiful, and salt only speeds rusting up rather than being a reactant that forms rust. So the oxygen-and-water equation is the right one.

20. (B) The white powder is magnesium oxide, a new substance formed when magnesium combined with oxygen; its water mixture behaving like a base (turning red litmus blue) is a property the original metal did not have, which confirms a chemical change. The ribbon was not merely melted, the result is not due to cleaning grit, and the bare metal does not itself turn litmus blue. The new property is the proof that a new substance was made.

21. (D) Camphor turning to vapour and then re-forming as solid on the cool funnel is sublimation followed by deposition of the same camphor, a physical change throughout. Naphthalene balls shrinking are also subliming directly to vapour, the same substance

leaving as a gas, which is physical too. No new substance forms in either case, so neither is chemical; vanishing completely does not make naphthalene chemical, and vaporising is not a chemical step. Both are physical.

22. (D) Giving off a gas does not always mean a chemical change: boiling water releases water vapour and dry ice subliming releases carbon dioxide gas, yet both are physical state changes, so that rule is false. The other three are sound: a chemical change is defined by new substances forming, freezing does release heat to the surroundings while staying physical, and crystallisation gives pure crystals of the same substance and is physical. So the gas rule is the false one.

23. (C) Where the paint chipped, air and moisture reached the exposed iron and rust formed; because rust is bulky and flaky it lifts and pushes off the surrounding paint, baring fresh iron that rusts in turn, which is why the spot spreads. The paint does not itself convert to rust, no solvent is corroding the iron, and intact paint protects rather than speeds rusting, so only the chip-exposure explanation is correct.

24. (C) A solid melting to a liquid of the same material and re-solidifying unchanged is a state change with no new substance, so it is the weakest evidence of a chemical change — indeed it points the other way. A new-smelling gas, a precipitate from mixing two clear liquids, and bursting into flame leaving a residue are all recognised clues that new substances formed. Therefore the simple melting is the weakest indicator of chemistry.

25. (D) Iron in the open solution can combine with oxygen and moisture from the air to form a brown rust-like substance at the surface, a chemical change that produces a new substance. The colour is not merely fading, since a new brown material is appearing; crystallisation would give the same pale solid, not a brown film; and dissolving further does not turn a solution brown. So the air-oxidation explanation is correct.

26. (D) A burning candle shows both at once: the solid wax melting and rising is a physical change, while the wax vapour burning to new substances at the flame is chemical. Ice melting and the drink cooling are both physical, salt spreading and the water turning salty are both physical (dissolving makes no new substance), and a wire bending with a little warming is entirely physical. So the candle is the event that genuinely combines a physical and a chemical change.

27. (C) Melting butter and letting it re-set is a state change of the same butter, a physical change, which shows that not every cooking step is chemical. Boiling an egg sets new substances, toasting bread browns it into new substances with a roasted smell, and caramelising sugar makes a new brown bitter substance — all chemical. So melting butter is the counter-example that disproves the over-broad claim.

28. (B) Stainless steel is an alloy and chrome plating is a surface layer; in each case a chromium-containing surface keeps the iron shielded from air and moisture, which is why

both resist rust. The iron is not turned into pure chromium, the protection works even when the items get wet, and speeding the iron's reaction with oxygen would corrode it, not protect it. So the shared protective-surface explanation is correct.

29. (D) Rust is heavy only in N1, the one nail that has both oxygen and moisture; every set-up missing either factor barely rusts, which shows both are needed together. If only oxygen mattered, the dry-air oxygen nail would have rusted heavily, and if only moisture mattered, the moist-air no-oxygen nail would have; neither did. So the experiment cleanly shows rusting requires both oxygen and moisture.

30. (B) A change is chemical exactly when a new substance forms; the direction of heat flow is irrelevant. The first two changes both form new substances, so both are chemical; the third and fourth form none, so both are physical regardless of the heat. That gives two chemical changes. Heat release is not required for chemistry, not every heat-involving change is chemical, and exchanging heat alone never makes a change chemical.

31. (B) Evaporating water from a salt solution recovers the same salt whether the heat is gentle sun or a strong flame, so both halves are physical changes; the quantity of heat used does not by itself create a new substance. Strong heat only drives off the water faster. So it is wrong to call the strongly heated half chemical, wrong to let the amount of heat decide the type, and wrong to claim evaporation makes a fresh compound.

32. (A) A chemical change is defined by the formation of a new substance, not by whether a dramatic signal happens to be visible; some reactions form new substances quietly with no obvious colour, gas, or flame. So Bina is right. Visibility is not the definition, the absence of a visible clue does not prove no new substance formed, and a clue-free change is not automatically physical. The deciding question is always whether a new substance was made.

33. (A) Both events have iron combining with oxygen to form new substances: rapidly with sparks in the flame, and slowly as rust in the damp salty air, where the salt speeds the corrosion. Speed does not decide the type, so slowness does not make rusting physical, and the sparks are burning iron, not mere hot bits. That iron remains in the bulk does not make either change physical. Hence both are chemical.

34. (B) Carbon dioxide from the air reacts slowly with limewater to form insoluble calcium carbonate, which appears as a faint white skin on the surface, a chemical change. The film is not dust left by evaporation, the surface has not frozen at ordinary room conditions, and oxygen does not bleach limewater white. So the reaction with carbon dioxide is the correct explanation.

35. (D) Melting solder and letting it set again are state changes of the same metal, which keeps its identity, so they are physical regardless of the substance being a metal. The heat

does not create a new substance, the cooled lump is the same solder not a new solid, and simple melting is not rusting. So melting a metal is a physical change.

36. (A) Diluting only spreads the same colouring matter through more water, so the colour looks weaker without any new substance forming; evaporating the water would bring the colour back, confirming a physical change. A colour change is not automatic proof of chemistry, water does not react with the dye here, and the change can be classified — it is physical. So no chemical change occurred.

37. (D) Magnetising and de-magnetising change only the arrangement that gives magnetism; the needle stays the same steel and no new substance forms, so both steps are physical. Gaining or losing magnetism is not the formation of a new substance, so neither is chemical, and merely heating the needle to de-magnetise it is not rusting. So both are physical changes.

38. (C) In both cups the salt simply dissolves, a physical change, and nothing new appears, so no reaction has occurred in either. Two dissolved substances do not automatically react, dissolved salt can be recovered by evaporation so it is not destroyed, and the mere presence of colour does not make one cup chemical. So both are physical dissolving with no reaction.

39. (B) Sea air is both humid and salty, and since salt speeds the rusting reaction the beachfront bench meets more moisture plus salt and rusts faster than the dry inland one. Cooler temperature is not what drives rusting, carbon dioxide is not the gas that rusts iron, and the difference is the salty damp environment, not a difference in the grade of iron. So the extra moisture and salt of sea air explain the faster corrosion.

40. (D) Failing to recover the original by a physical means is strong evidence that a new substance formed, so Test 1 is the best indicator of a chemical change here. A small temperature rise alone does not prove chemistry, the absence of a colour change proves nothing about being chemical, and seeing no gas certainly does not prove a new substance formed. So the recovery failure is the strongest single piece of evidence.

41. (D) Ice melting is only a state change of the same water, so it is physical; bread going mouldy involves mould growth that produces new substances and a new smell, so it is chemical. Being left in the open does not make both physical, warmth from the surroundings does not make both chemical, and absorbing heat does not make the melting chemical. So the melting is physical and the mould is chemical.

42. (A) Crystallisation purifies physically: the soluble copper sulphate dissolves so that insoluble impurities can be filtered out, and as crystals grow they exclude soluble impurities, which remain in the leftover solution. The impurities are not converted into copper sulphate, the dissolving heat does not burn impurities away, and the crystals are

the same copper sulphate, not a new compound. So the dissolve-filter-crystallise mechanism explains the purification.

43. (A) Inability to recover the starting materials by a physical step is the most decisive sign that a new substance formed, and the limewater going milky confirms the gas is carbon dioxide produced by the new chemistry. A gas alone is not proof, since physical changes such as boiling release gas too; limewater milkiness is a specific test for carbon dioxide rather than a general proof of any chemistry; and measuring heat is not required to confirm a chemical change. So the recovery failure, backed by the gas test, is the decisive reason.

44. (C) Oil forms a barrier that keeps air and moisture from the iron, so the oiled hinge is protected, not at risk; only the bare bracket, exposed to both, rusts heavily. Oil does not draw in moisture to speed rusting, glossy paint protects rather than accelerates rusting, and the galvanised pail's zinc protects the iron rather than leaving it to rust faster than bare metal. So the student's prediction about the oiled hinge is wrong.

45. (B) Fermentation of dough makes new substances (the gas and flavours that raise and sour it), so it is chemical, while sharpening a pencil only shaves off shape, a physical change; this pair is fully correct. In the others, the coating step of galvanising is a physical layering while cooking an egg is chemical (the reverse of what is claimed), naphthalene subliming is physical while rusting is chemical, and dissolving washing soda is physical while digestion is chemical. So only the fermentation-and-sharpening pair classifies both members correctly.

46. (A) The reaction makes a new substance, a gas that turns limewater milky and does not support burning, both of which identify carbon dioxide, so a chemical change occurred. The fizzing is not escaping dissolved air, the gas is not oxygen since it fails to support a flame, and it is not hydrogen since the limewater test points to carbon dioxide. So the change is chemical and the gas is carbon dioxide.

47. (A) Both kinds of change can take in or give out energy: freezing (physical) releases heat, melting (physical) absorbs it, and chemical changes can do either too, so energy flow alone cannot tell physical from chemical. It is false that only chemical changes release energy, false that chemical changes always release it, and false that the direction of energy flow sorts changes into types. So energy flow alone is not decisive.

48. (C) Crystallisation simply brings the same sugar out of solution as crystals, a physical change, while the browning at the overheated edge turns sugar into a new dark substance, a chemical change. Separating from water is not chemical, the crystals are the same sugar not a new substance, so it is wrong to call crystallisation chemical. So crystal-forming is physical and the browning is chemical.

49. (B) Both coatings keep air and moisture off the iron, which is their shared protective role, but zinc is more reactive than iron and so keeps protecting even at a scratch, whereas tin does not, so they are not identical. Neither turns iron into a new metal, they do not work in completely opposite ways (both seal the surface), and the coatings, not the food, are what protect the iron. So the reply noting both the shared sealing and zinc's extra sacrificial action is most accurate.

50. (B) Rusting, burning, and respiration are all chemical changes involving oxygen: burning is rapid, respiration is a slow controlled use of oxygen inside the body that releases energy, and rusting is slow oxidation at the iron's surface. All three make new substances, so it is wrong to call any of them physical; respiration in particular forms new substances and is chemical, not merely warmth. So all three are chemical.

51. (C) Dissolving salt and evaporating to get the same salt back is reversible and keeps the same substance throughout, so it supports the idea that reversibility goes with physical changes. Burning a candle, cooking an egg, and rusting a nail are all chemical and cannot be reversed by simple physical means, so they do not support the claim. So the salt example is the right support.

52. (A) Being irreversible does not by itself make a change chemical; the egg is chemical because cooking forms new substances, whereas the smashed tile, though irreversible, is still the same clay and so is physical. So the egg verdict is right but for the right reason, not irreversibility. It is wrong to say irreversibility guarantees chemistry, wrong to call the egg physical, and wrong to say the egg forms no new substance. So the flaw is treating irreversibility as proof.

53. (B) The boiled-water nail had water but no air and did not rust, which shows water by itself is not enough; only the nail with both water and air rusted, so both are needed. The dry-air nail had air but no water and did not rust, so air alone is not enough either, and the open-water nail rusted because of the conditions, not because it was disturbed. So rusting needs both water and air.

54. (A) Dissolving sugar is physical, toasting bread to brown forms new substances and is chemical, and bending a spoon is physical, so this list is physical-chemical-physical as required. The others fail the pattern: rusting is chemical not physical to start; freezing, crystallising and sharpening are all physical (no chemical middle); and burning a sparkler is chemical not physical to start. So only the sugar-toast-spoon list fits the required order.

55. (A) Melting wax and the same liquid wax rising up the wick are both physical, since the wax keeps its identity; only the wax vapour burning at the flame forms new substances and is chemical. So two are physical and one is chemical. The flame's heat does not make melting and rising chemical, the wax does not stay unchanged forever since it burns, and climbing the wick does not change the wax's identity. So the count is two physical and one chemical.

56. (B) Heating ice to melt it needs heat yet only changes the state of the same water, which freezes back unchanged, so it is physical and directly refutes the rule that any heat-needing change is chemical. Caramelising sugar, heating limestone, and burning magnesium all need heat and form new substances, so they are chemical and support rather than refute the claim. So melting ice is the refuting example.

57. (B) The black layer is a new copper-oxygen compound formed where copper combined with oxygen from the air, a chemical surface change, shown by the fresh copper beneath when scraped. It is not soot merely sitting on top, not just the colour of hot copper, and not melted-and-refrozen copper. So a new copper-oxygen compound is the correct explanation.

58. (D) Both halves recover the same salt, which the flame-dried solid is confirmed to be, so evaporating the water is physical regardless of how fierce the heat was; the strength of heating does not by itself create a new substance. So calling the flame-dried half chemical is wrong, the heat strength does not decide the type, and evaporation does not build a fresh compound. Both are physical.

59. (D) Crystallisation gives back the same substance as pure crystals with no new substance, so it is physical, while rusting forms a new substance from iron, oxygen, and water, so it is chemical. They are therefore not both chemical and not both physical, and the roles are not reversed — crystallisation is the physical one and rusting the chemical one. So the contrast as stated is correct.

60. (A) Rust is iron joined with oxygen and water, so the iron that converts to rust picks up the mass of that oxygen and water; that is why 12 g of iron lost from the sheet ends up as more than 12 g of flakes, here 21 g. The extra is added oxygen and water, not trapped air bubbles and not matter made from nothing, and the steady thinning of the solid sheet is normal rusting rather than a sign of impurity. So the flakes outweigh the iron consumed because rust is heavier than the iron it formed from.